Ideation Plan: Nadiaris Labs & AMD Vitis AI Integration

Project Vision: Building a Hybrid-AI Cybersecurity Auditor that combines local hardware speed (AMD Vitis AI) with cloud intelligence (Google Gemini).

1. The Core Innovation: "The Hybrid Guard"

Most cybersecurity startups rely 100% on cloud APIs. This creates latency, costs money, and exposes sensitive code to the internet. **Our Innovation:** Use AMD Vitis AI to perform a "First-Pass" scan locally on the hardware.

- **Tier 1 (Local):** Instant detection of known patterns (Vitis AI + ONNX).
- **Tier 2 (Cloud):** Deep semantic reasoning for unknown threats (Gemini).

2. Technical Roadmap: From Prototype to Product

Phase A: Hardware Optimization (The Vitis Shift)

- **Model Selection:** Use the ONNX models for Phishing and Malware detection.
- **Quantization Strategy:** Convert FP32 models to INT8 using the `vai_q_onnx` tool to fit the DPU (Deep Learning Processor Unit).
- **Execution Provider:** Update `ml_engine/inference.py` to prioritize `VitisAIExecutionProvider`.
- **Hardware Target:** Optimize for AMD Ryzen AI (NPU) or Kria/Alveo (FPGA) targets.

Phase B: Smart Orchestration Logic

Update the FastAPI `main.py` to handle the new decision tree:

1. **Incoming Request:** Code snippet arrives.
2. **Local Inference:** Run Vitis AI scan ($< 1ms$).
3. **Threshold Check:** * If Confidence $> 95\%$: Return local result immediately (Fastest path).
   - If Confidence $< 95\%$: Send to Gemini for "Second Opinion" (Expert path).
4. **Sync:** Log both paths to the PostgreSQL "Vault."

Phase C: Competitive Differentiation (The "Pitch" Points)

- **Data Sovereignty:** "Your code stays on your machine unless an anomaly is found."
- **Zero-Latency Shield:** Real-time protection that doesn't wait for the internet.
- **Sustainability:** Reducing cloud compute carbon footprint by offloading 90% of tasks to low-

power AMD hardware.

3. Hackathon Deliverables Checklist

1. The "Vitis AI" Demo

Show a side-by-side comparison:

- **Cloud-only scan:** 7 seconds.
- **Vitis AI-accelerated scan:** 1 millisecond.
- **Visual:** A dashboard showing "Hardware Offloading" active.

2. The Multi-Layered Logs

Show the PostgreSQL database tracking which "Provider" caught the threat:

- `provider_used: VitisAI_DPU`
- `provider_used: Gemini_Flash`

3. The Deployment Story

Document how the project can be deployed as:

- An **Enterprise Appliance** (FPGA-based rack).
- A **Developer Tool** (Ryzen AI laptop plugin).

4. Key Questions to Answer for Judges

- **How does Vitis AI help?** It eliminates the "speed-vs-intelligence" trade-off.
- **Is it scalable?** Yes, because the heavy lifting is done by the user's local AMD hardware, not our servers.
- **What is the business model?** SaaS for cloud audits, plus a "Hardware-Accelerated" license for local/private environments.

5. Immediate Next Steps (The 24-Hour Sprint)

1. **Test Quantization:** Run your ONNX model through the Vitis AI quantizer.
2. **Integrate `VitisAISession`:** Replace the dummy local model in your project with the Vitis AI provider.
3. **Update Database:** Ensure the `provider_used` column in `ThreatLog` correctly identifies hardware-based hits.